

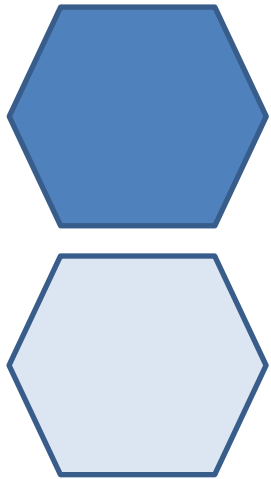


# Bioinformatics

**Ing. Giulia Fiscon, PhD**

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Department of Computer, Control, and Management Engineering  
Sapienza University of Rome



# NCBI - RefSeq

Reference Sequences

<http://www.ncbi.nlm.nih.gov/refseq/>

# RefSeq

- The Reference Sequence (RefSeq) database is a **curated collection of DNA, RNA, and protein sequences** built by NCBI
- Unlike GenBank, RefSeq **provides only one example of each natural biological molecule** for major organisms ranging from viruses to bacteria to eukaryotes
- For each model organism, RefSeq aims to provide **separated and linked records** for the genomic DNA, the gene transcripts, and the proteins arising from those transcripts
- To produce RefSeq records, NCBI selects the best available information on each molecule and updates the records as more information emerges

# RefSeq

- A commonly used analogy is that if GenBank is like the primary research literature, RefSeq is like the review literature
- In some cases, creation of a RefSeq record involves no more than selecting a single good example from GenBank and making a copy in RefSeq
- In other cases, NCBI in-house staff generates and annotates the records based on the existing primary data, sometimes by combining parts of several GenBank records
- Also, some records are automatically imported from other curated databases, such as the SGD database of yeast genome data and the FlyBase database of Drosophila genomes
- The approach selected for creating a RefSeq record depends on the specific organism and the quality of available information

# RefSeq

- When NCBI first creates a RefSeq record, the record initially reflects only the information from the source GenBank record with added links
- At this point, the record has not yet been reviewed by NCBI staff, and therefore it is identified as “**provisional**”
- After NCBI examines the record – often adding information from other GenBank records, such as the sequences for the 5’UTR and 3’UTR, and providing further literature references – it is marked as “**reviewed**”

# RefSeq record

- RefSeq records are displayed in a format similar to that of GenBank records from which they are derived
- However, they can be distinguished from GenBank records by their accession prefix, which includes an underscore, and a notation in the “comment” field that indicates the RefSeq status
- RefSeq records can be accessed through NCBI’s Nucleotide and Protein databases, which are among the many databases linked through the retrieval system
- When retrieving search results, users can choose to see all GenBank records or only RefSeq records by clicking on the appropriate tab at the top of the results page



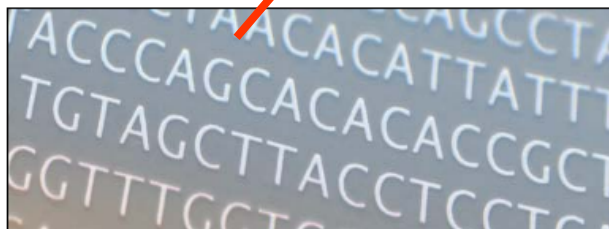
Nucleotide

Nucleotide ▼

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## Nucleotide

The Nucleotide database is a collection of sequences from several sources, including GenBank, RefSeq, TPA and PDB. Genome, gene and transcript sequence data provide the foundation for biomedical research and discovery.

### Using Nucleotide

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[GenBank FTP](#)

[RefSeq FTP](#)

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[INSDC](#)



RefSeq

RefSeq ▼

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## RefSeq: NCBI Reference Sequence Database

A comprehensive, integrated, non-redundant, well-annotated set of reference sequences including genomic, transcript, and protein.

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[Human Reference Genome](#)

[Prokaryotic RefSeq Genomes](#)

[FAQ](#)

[NCBI Handbook](#)

[Factsheet](#)

### RefSeq Access

[Human Genome Resources and Download](#)

[RefSeq FTP](#)

[RefSeq genomes FTP](#)

[New RefSeq genomic \(last 30 days\)](#)

[New RefSeq transcripts \(last 30 days\)](#)

[New RefSeq proteins \(last 30 days\)](#)

[Searching for RefSeq records \(Queries\)](#)

### RefSeq projects

[Consensus CDS \(CCDS\)](#)

[RefSeq Functional Elements](#)

[RefSeqGene](#)

[Targeted Loci](#)

[Virus Variation](#)

[RefSeq Select](#)

[MANE](#)

### Announcements

September 16, 2022

**RefSeq Release 214 is available for FTP**

This release includes:

Proteins: 239,609,016

Transcripts: 47,387,931

Organisms: 123,394

Available at: <ftp://ftp.ncbi.nlm.nih.gov/refseq/release/>

Documentation: [Release Notes](#)

See [previous announcements](#), follow [NCBI on Twitter](#), or subscribe to [NCBI's refseq-announce mail list](#) to receive announcements.

### Related Links

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[Gene](#)

[Genome](#)

[Genome Data Viewer](#)

[Annotated Eukaryotic Genomes](#)

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# RefSeq FAQ

## RefSeq Frequently Asked Questions (FAQ)

Kim Pruitt, Ph.D., Garth Brown, Ph.D., and Mike Murphy, Ph.D.

► [Author Information](#)

Created: November 15, 2010; Last Update: May 31, 2016.

### General Information about RefSeq

[What is a Reference Sequence \(RefSeq\)?](#)

[What are the distinguishing features of a RefSeq record?](#)

[How are RefSeq records provided?](#)

[Where can I find more information about the NCBI RefSeq project?](#)

[How do I access RefSeq records?](#)

[What is the difference between XM\\_ and NM\\_ accessions?](#)



[What is an NG\\_ accession and why are they made?](#)

[What sequence is used to define a RefSeq?](#)

[What is the difference between RefSeq and GenBank?](#)

[The sequence of a RefSeq accession is identical to that of a GenBank accession. Will one be removed?](#)

[How can I quickly identify RefSeq records?](#)

[If RefSeq is a non-redundant database why does my BLAST query return hits to more than one RefSeq accession?](#)

[How do I cite the RefSeq project, a species-specific dataset, or individual RefSeq records?](#)

[How can I evaluate the support for a RefSeq transcript or protein?](#)

[Why aren't RefSeq records made for all organisms or for all of the loci available in Gene?](#)

# XM\_ vs NM\_ accessions

## What is the difference between XM\_ and NM\_ accessions?

Accession numbers that begin with the prefix XM\_ (mRNA), XR\_ (non-coding RNA), and XP\_ (protein) are model RefSeqs produced either by NCBI's genome annotation pipeline or copied from computationally annotated submissions to the INSDC. These RefSeq records are derived from the genome sequence and have varying levels of transcript or protein homology support. They represent the predicted transcripts and proteins annotated on the NCBI RefSeq contigs and may differ from INSDC mRNA submissions or from the subsequently curated RefSeq records (with NM\_, NR\_, or NP\_ accession prefixes). These differences may reflect real sequence variation (polymorphism), or errors or gaps in the available genome sequence. The support for model RefSeq records should be further evaluated by comparing them to other sequence information available in Gene, BLink, Related Sequences, and BLAST reports.

The genome annotation pipelines are automated and their predicted products may or may not be subject to manual curation, but the data may be refreshed periodically.

A complete description of RefSeq accessions prefixes can be found [here](#).



Table 1. [RefSeq](#) accession numbers and molecule types.

| Accession prefix | Molecule type        | Comment                                               |
|------------------|----------------------|-------------------------------------------------------|
| AC_              | Genomic              | Complete genomic molecule, usually alternate assembly |
| NC_              | Genomic              | Complete genomic molecule, usually reference assembly |
| NG_              | Genomic              | Incomplete genomic region                             |
| NT_              | Genomic              | Contig or scaffold, clone-based or WGS <sup>a</sup>   |
| NW_              | Genomic              | Contig or scaffold, primarily WGS <sup>a</sup>        |
| NS_              | Genomic              | Environmental sequence                                |
| NZ_ <sup>b</sup> | Genomic              | Unfinished WGS                                        |
| NM_              | <a href="#">mRNA</a> |                                                       |
| NR_              | <a href="#">RNA</a>  |                                                       |
| XM_ <sup>c</sup> | <a href="#">mRNA</a> | Predicted model                                       |
| XR_ <sup>c</sup> | <a href="#">RNA</a>  | Predicted model                                       |
| AP_              | Protein              | Annotated on AC_ alternate assembly                   |
| NP_              | Protein              | Associated with an NM_ or NC_ accession               |
| YP_ <sup>c</sup> | Protein              |                                                       |
| XP_ <sup>c</sup> | Protein              | Predicted model, associated with an XM_ accession     |
| ZP_ <sup>c</sup> | Protein              | Predicted model, annotated on NZ_ genomic records     |

<sup>a</sup> Whole Genome Shotgun sequence data.

<sup>b</sup> An ordered collection of [WGS sequence](#) for a genome.

<sup>c</sup> Computed.

# ReSeq Accession numbers

## mRNAs and Proteins

NM\_123456

Curated mRNA

NP\_123456

Curated Protein

NR\_123456

Curated non-coding RNA

XM\_123456

Predicted mRNA

XP\_123456

Predicted Protein

XR\_123456

Predicted non-coding RNA

## Gene Records

NG\_123456

Reference Genomic Sequence

## Chromosome

NC\_123455

Microbial replicons, organelle genomes,  
human chromosomes

## Assemblies

NT\_123456

Contig

NW\_123456

WGS Supercontig



RefSeq

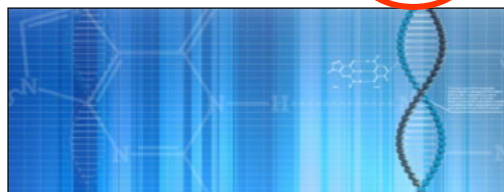
RefSeq



TP53



Search



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A comprehensive, integrated, non-redundant, well-annotated set of reference sequences including genomic, transcript, and protein.

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[Annotated Eukaryotic Genomes](#)

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[Collaborators](#)



Nucleotide

Nucleotide



Search

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Species

Animals (9,391)  
Plants (50)  
Fungi (35)  
Protists (17)  
Customize ...

Molecule types

genomic DNA/RNA (3,518)  
mRNA (5,737)  
Customize ...

Source databases

RefSeq (9,493)  
Customize ...

Sequence Type

Nucleotide (9,493)

Sequence length

Custom range...

Release date

Custom range...

Revision date

Custom range...

[Clear all](#)

[Show additional filters](#)

Summary ▾ 20 per page ▾ Sort by Default order ▾

Send to: ▾

Filters: [Manage Filters](#)

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☐ [Homo sapiens tumor protein p53 \(TP53\), RefSeqGene \(LRG\\_321\) on chromosome 17](#)

1. 22,772 bp linear DNA

Accession: NG\_017013.2 GI: 383209646

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 10, mRNA](#)

2. 2,522 bp linear mRNA

Accession: NM\_001407264.1 GI: 2246031143

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 9, mRNA](#)

3. 2,615 bp linear mRNA

Accession: NM\_001407263.1 GI: 2246031136

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 13, mRNA](#)

4. 2,658 bp linear mRNA

Accession: NM\_001407271.1 GI: 2246031099

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

Results by taxon

Top Organisms [\[Tree\]](#)

Homo sapiens (556)  
Mus musculus (162)  
Danio rerio (86)  
Rattus norvegicus (63)  
Gallus gallus (49)  
All other taxa (8577)

[More...](#)

Find related data

Database:

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Search details

tp53[All Fields] AND srcdb\_refseq[PROP]

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Nucleotide

Nucleotide

tp53 AND srcdb\_refseq[PROP]



Search

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Species

Animals (9,391)

Plants (50)

Fungi (35)

Protists (17)

Customize ...

Molecule types

genomic DNA/RNA (3,518)

mRNA (5,737)

Customize ...

Source databases

RefSeq (9,493)

Customize ...

Sequence Type

Nucleotide (9,493)

Sequence length

Custom range...

Release date

Custom range...

Revision date

Custom range...

Clear all

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Items: 1 to 20 of 9493

<< First < Prev Page 1 of 475 Next > Last >>

☐ [Homo sapiens tumor protein p53 \(TP53\), RefSeqGene \(LRG\\_321\) on chromosome 17](#)

1. 32,772 bp linear DNA

Accession: NG\_017013.2 GI: 383209646

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 10, mRNA](#)

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Accession: NM\_001407264.1 GI: 2246031143

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

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[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

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Accession: NM\_001407271.1 GI: 2246031099

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

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Top Organisms [Tree](#)

Homo sapiens (556)

Mus musculus (162)

Danio rerio (86)

Rattus norvegicus (63)

Gallus gallus (49)

All other taxa (8577)

More...

Find related data

Database: [Select](#)

Find items

Search details

tp53[All Fields] AND srcdb\_refseq[PROP]

Search

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Nucleotide

Nucleotide

(tp53 AND srcdb\_refseq[PROP]) AND "Homo sapiens"[porgn: \_\_txid9606]



Search

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Species

Animals (556)

Customize ...

Molecule types

genomic DNA/RNA (33)

mRNA (451)

Customize ...

Source databases

RefSeq (556)

Customize

Sequence Type

Nucleotide (556)

Sequence length

Custom range...

Release date

Custom range...

Revision date

Custom range...

Clear all

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Summary 20 per page Sort by Default order

Send to:

Filters: [Manage Filters](#)

Items: 1 to 20 of 556

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☐ [Homo sapiens tumor protein p53 \(TP53\), RefSeqGene \(LRG\\_321\) on chromosome 17](#)

1. 32,772 bp linear DNA  
Accession: NG\_017013.2 GI: 383209646  
[Protein](#) [PubMed](#) [Taxonomy](#)  
[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 10, mRNA](#)

2. 2,522 bp linear mRNA  
Accession: NM\_001407264.1 GI: 2246031143  
[Protein](#) [PubMed](#) [Taxonomy](#)  
[GenBank](#) [FASTA](#) [Graphics](#)

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Accession: NM\_001407263.1 GI: 2246031136  
[Protein](#) [PubMed](#) [Taxonomy](#)  
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☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 13, mRNA](#)

4. 2,658 bp linear mRNA  
Accession: NM\_001407271.1 GI: 2246031099  
[Protein](#) [PubMed](#) [Taxonomy](#)  
[GenBank](#) [FASTA](#) [Graphics](#)

Find related data

Database: [Select](#)

Find items

Search details

(tp53[All Fields] AND  
srcdb\_refseq[PROP]) AND "Homo sapiens"  
[porgn]

Search

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(tp53 AND srcdb\_refseq[PROP]) AND  
"Homo sapiens"[porgn] (556) Nucleotide

tp53 AND srcdb\_refseq[PROP] (9493) Nucleotide

ATP synthase subunit 8 (mitochondrion)  
[Ciona robusta] Protein

TP53 tumor protein p53 [Canis lupus  
familiaris] Gene

tp53 canis AND (alive[prop]) (34)





Nucleotide

Nucleotide

Advanced

GenBank

Send to:

## Homo sapiens tumor protein p53 (TP53), RefSeqGene (LRG\_321) on chromosome 17

NCBI Reference Sequence: NG\_017013.2

[FASTA](#) [Graphics](#)

Go to:

LOCUS NG\_017013 32772 bp DNA linear PRI 25-SEP-2022  
DEFINITION Homo sapiens tumor protein p53 (TP53), RefSeqGene (LRG\_321) on chromosome 17.  
ACCESSION NG\_017013  
VERSION NG\_017013.2  
KEYWORDS RefSeq; RefSeqGene.  
SOURCE Homo sapiens (human)  
ORGANISM [Homo sapiens](#)  
Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Primates; Haplorrhini; Catarrhini; Hominidae; Homo.  
REFERENCE 1 (bases 1 to 32772)  
AUTHORS Mirgayazova R, Khadiullina R, Chasov V, Mingaleeva R, Miftakhova R, Rizvanov A and Bulatov E.  
TITLE Therapeutic Editing of the TP53 Gene: Is CRISPR/Cas9 an Option?  
JOURNAL Genes (Basel) 11 (6), E704 (2020)  
PUBMED [32630614](#)  
REMARK Review article  
Publication Status: Online-Only  
REFERENCE 2 (bases 1 to 32772)  
AUTHORS Flynt E, Bisht K, Sridharan V, Ortiz M, Towfic F and Thakurta A.  
TITLE Prognosis, Biology, and Targeting of TP53 Dysregulation in Multiple Myeloma  
JOURNAL Cells 9 (2), E287 (2020)

Scrolling down



JOURNAL EMBO J 3 (13), 3257-3262 (1984)  
PUBMED [6396087](#)  
COMMENT REVIEWED [REFSEQ](#): This record has been curated by NCBI staff in collaboration with Magali Olivier, Thierry Soussi, Jean-Christophe Bourdon, Nazneen Rahman. The reference sequence was derived from [AC087388.9](#) and [AC007421.13](#).  
This sequence is a reference standard in the [RefSeqGene](#) project. On Apr 3, 2012 this sequence version replaced [NC\\_017013.1](#).

Summary: This gene encodes a tumor suppressor protein containing transcriptional activation, DNA binding, and oligomerization domains. The encoded protein responds to diverse cellular stresses to regulate expression of target genes, thereby inducing cell cycle arrest, apoptosis, senescence, DNA repair, or changes in metabolism. Mutations in this gene are associated with a variety of human cancers, including hereditary cancers such as Li-Fraumeni syndrome. Alternative splicing of this gene and the use of alternate promoters result in multiple transcript variants and isoforms. Additional isoforms have also been shown to result from the use of alternate translation initiation codons from identical transcript variants (PMIDs: 12032546, 20937277). [provided by RefSeq, Dec 2016].

COMPLETENESS: complete on the 3' end.

# RefSeq record

- When retrieving search results, users can choose to see all GenBank records or only RefSeq records by clicking on the appropriate tab at the top of the results page
- We search again for TP53 in Homo Sapiens, but first searching in all GenBank records, then selecting RefSeq from the left side menu in order to narrow down the search field only to RefSeq records

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National Center for Biotechnology Information

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## Nucleotide

The Nucleotide database is a collection of sequences from several sources, including GenBank, RefSeq, TPA and PDB. Genome, gene and transcript sequence data provide the foundation for biomedical research and discovery.

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### Nucleotide Tools

### Other Resources

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#### Species

Animals (14,201)  
Plants (130)  
Fungi (299)  
Protists (73)  
Bacteria (56)  
Archaea (21)  
Viruses (31)  
Customize ...

#### Molecule types

genomic DNA/RNA (7,304)  
mRNA (7,316)  
Customize ...

#### Source databases

INSDC (GenBank) (6,019)  
RefSeq (9,493)  
Customize ...

#### Sequence Type

Nucleotide (14,331)  
EST (1,202)  
GSS (3)

#### Genetic compartments

Mitochondrion (2)

#### Sequence length

Custom range...

#### Release date

Custom range...

#### Revision date

Custom range...

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GENE

Was this helpful?

#### TP53 – tumor protein p53

[Homo sapiens \(human\)](#)

Also known as: BCC7, BMFS5, LFS1, P53, TRP53

Gene ID: 7157

[RefSeq transcripts \(26\)](#) [RefSeq proteins \(25\)](#) [RefSeqGene \(1\)](#) [PubMed \(11,212\)](#)

#### RefSeq Sequences

Items: 1 to 20 of 15536

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☐ [Homo sapiens TP53 \(TP53\) gene, exon 5 and partial cds](#)

1. 123 bp linear DNA  
Accession: MH011443.1 GI: 1486783306  
[Protein](#) [Taxonomy](#)  
[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens isolate 88F TP53 \(TP53\) gene, exon 8 and partial cds](#)

2. 57 bp linear DNA

#### Results by taxon

Top Organisms [\[Tree\]](#)

[Homo sapiens \(4164\)](#)  
[synthetic construct \(636\)](#)  
[Mus musculus \(186\)](#)  
[Danio rerio \(90\)](#)  
[Rattus norvegicus \(87\)](#)  
[All other taxa \(10373\)](#)  
[More...](#)

#### Find related data

Database:

#### Search details

TP53[All Fields]

[See more...](#)

#### Recent activity

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TP53 (15536)

TP53 AND (alive[prop]) (8481)

Click on Homo sapiens



Nucleotide

Nucleotide

(TP53) AND "Homo sapiens"[porgn: \_\_txid9606]



Search

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Species

Animals (4,164)

Customize ...

Molecule types

genomic DNA/RNA (2,288)

mRNA (1,795)

Customize ...

Source databases

INSDC (GenBank) (3,600)

RefSeq (556)

Customize ...

Sequence Type

Nucleotide (3,014)

EST (1,148)

GSS (2)

Sequence length

Custom range...

Release date

Custom range...

Revision date

Custom range...

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☐ [Homo sapiens TP53 \(TP53\).gene.exon 5 and partial cds](#)

1. 123 bp linear DNA

Accession: MH366483.1 GI: 1486783306

[Protein](#)

[Taxonomy](#)

[Graphics](#)

[Homo sapiens isolate 88F TP53 \(TP53\).gene.exon 8 and partial cds](#)

2. 57 bp linear DNA

Accession: MH366483.1 GI: 1472901973

[Protein](#)

[Taxonomy](#)

[GenBank](#)

[FASTA](#)

[Graphics](#)

[PopSet](#)

☐ [Homo sapiens isolate 6F TP53 \(TP53\).gene.exon 8 and partial cds](#)

3. 57 bp linear DNA

Accession: MH366482.1 GI: 1472901971

[Protein](#)

[Taxonomy](#)

[GenBank](#)

[FASTA](#)

[Graphics](#)

[PopSet](#)

☐ [Homo sapiens isolate 8F TP53 \(TP53\).gene.exon 8 and partial cds](#)

4. 63 bp linear DNA

Accession: MH366481.1 GI: 1472901969

[Protein](#)

[Taxonomy](#)

[GenBank](#)

[FASTA](#)

[Graphics](#)

[PopSet](#)

☐ [Homo sapiens isolate 16F TP53 \(TP53\).gene.exon 8 and partial cds](#)

5. 66 bp linear DNA

Accession: MH366480.1 GI: 1472901967

Find related data

Database: [Select](#)

[Find items](#)

Search details

TP53[All Fields] AND "Homo sapiens"  
[porgn]

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[Q](#) (TP53) AND "Homo sapiens"[porgn] (4164)  
Nucleotide

[Q](#) TP53 (15536)  
Nucleotide

[Q](#) TP53 AND (alive[prop]) (8481)  
Gene

[Q](#) Homo sapiens tumor protein p53 (TP53),  
RefSeqGene (LRG\_321) on chromo Nucleotide

[Q](#) (tp53 AND srcdb\_refseq[PROP]) AND  
"Homo sapiens"[porgn] (556)  
Nucleotide

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Click on  
RefSeq



Nucleotide

Nucleotide

(TP53) AND "Homo sapiens"[porgn:\_\_txid9606]

Search

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Help

Species

Animals (556)

Customize ...

Molecule types

genomic DNA/RNA (33)

mRNA (451)

Customize ...

Source databases

INSDC (GenBank ...)

✓ RefSeq (556)

Customize ...

Sequence Type

Nucleotide (556)

Sequence length

Custom range...

Release date

Custom range...

Revision date

Custom range...

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Summary 20 per page Sort by Default order

Items: 1 to 20 of 556

Filters activated: RefSeq. Clear all

☐ [Homo sapiens tumor protein p53 \(TP53\), RefSeqGene \(LRG\\_321\) on chromosome 17](#)

1. 32,772 bp linear DNA

Accession: NG\_017013.2 GI: 383209646

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 10, mRNA](#)

2. 2,522 bp linear mRNA

Accession: NM\_001407264.1 GI: 2246031143

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 9, mRNA](#)

3. 2,615 bp linear mRNA

Accession: NM\_001407263.1 GI: 2246031136

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 13, mRNA](#)

4. 2,658 bp linear mRNA

Accession: NM\_001407271.1 GI: 2246031099

[Protein](#) [PubMed](#) [Taxonomy](#)

[GenBank](#) [FASTA](#) [Graphics](#)

☐ [Homo sapiens tumor protein p53 \(TP53\), transcript variant 13, mRNA](#)

5. 2,615 bp linear mRNA

We obtain the same results  
of the previous query

Filters: [Manage Filters](#)

Find related data

Database: Select

Find items

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TP53[All Fields] AND "Homo sapiens"  
[porgn] AND refseq[filter]

Search

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(TP53) AND "Homo sapiens"[porgn] AND  
(refseq[filter]) (556) Nucleotide

(TP53) AND "Homo sapiens"[porgn] (4164)  
Nucleotide

TP53 (15536)  
Nucleotide

TP53 AND (alive[prop]) (8481)  
Gene

Homo sapiens tumor protein p53 (TP53),  
RefSeqGene (LRG\_321) on chromo Nucleotide

See more...

# GenBank *versus* RefSeq

## GenBank

- Not curated
- Author can submit
- Only the author can revise multiple records for same common loci
- Records can contradict each other
- No limit to species that are included
- Data exchanged among INSDC members
- Similar to primary literature
- Proteins identified and linked
- Access via NCBI Nucleotide database

## RefSeq

- Curated
- NCBI creates from existing data
- NCBI revises as new data emerge
- Single records for each molecule of major organisms
- Limited to model organisms
- Exclusive NCBI database
- Similar to review articles
- Proteins and transcripts identified and linked
- Access via NCBI Nucleotide & Protein databases